

School of Computing and Information Technology

Database Design CMP2018 (Practical)

Group Project – PART 2 (LAB BETA)

General Instructions

1. This lab is an individual assessment.
Although the ERD was completed in groups, each student must complete and submit their own SQL implementation of the assigned scenario. The ERD should guide your table structure, including primary and foreign key relationships.
2. You are expected to demonstrate your understanding of data integrity rules through the proper use of constraints.
3. Your SQL script must execute without errors in your SQL environment (SQLSERVER, MySQL, PostgreSQL, or Oracle).
4. Include comments (--) to label each section of your script.

Task 1: Write SQL queries using aggregate functions to summarize data from your database (3 x 5 marks)

Scenario – Greg's Extreme Fitness Centre

- i) Calculate the total amount paid by all members for their fitness plans.
- ii) Count how many members are currently enrolled at the gym.
- iii) Find the lowest membership fee among all available plans.
- iv) Find the maximum BMI recorded among members.
- v) Determine the average monthly payment received from members

Scenario – Elegant Events

- i) Calculate the total payment made to all vendors for a specific event.
- ii) Count the number of events booked by each client.
- iii) Find the lowest vendor service cost recorded.
- iv) Identify the most expensive event budget in the system.
- v) Determine the average number of guests per event.

Scenario – Zoo Jam

- i) Find the total amount spent on animal feed for all enclosures.
- ii) Count the number of animals assigned to each caretaker.
- iii) Determine the minimum temperature required for any animal's habitat.
- iv) Find the oldest animal age recorded in the zoo.

- v) Calculate the average feeding cost per animal.

Prepared by: D. Allen Semester 1 AY 2025-26

Scenario 4 – Collections Museum

- i) Calculate the total insurance value of all artifacts in the museum.
- ii) Count the number of artifacts in each collection or exhibit.
- iii) Identify the oldest artifact based on year of origin.
- iv) Find the highest restoration cost recorded for any artifact.
- v) Compute the average display duration (in days) for artifacts on loan.

Scenario 5 – Football Federation

- i) Calculate the total number of goals scored by each team in the league.
- ii) Count the number of players registered under each club.
- iii) Find the youngest player's age in the database.
- iv) Identify the highest player salary in the federation.
- v) Determine the average number of matches played per team.

Scenario 6 – Promaneng Co. Ltd (Construction Projects)

- i) Calculate the total project cost for all ongoing construction projects.
- ii) COUNT – Count how many employees are assigned to each project.
- iii) MIN – Find the minimum contract value among the company's projects.
- iv) MAX – Identify the highest payment made to a subcontractor.
- v) AVG – Determine the average completion percentage of all active projects.

Task 2: Performing simple queries (9 + 6 marks)

- i) **Pattern matching** – Write a query to demonstrate pattern matching using a) % and b) checking the value of the third character in a field.
- ii) **Sorting** – Display information from a table sorted on multiple fields using both ascending and descending order in the query (e.g. last name within gender)
- iii) **Calculated Fields** – Choose the query most applicable to you
 - a) Display all members and their BMI whose BMI falls between 18.5 and 25. Calculate the **BMI** for each member using the formula $weight / (height * height)$.
 - b) Display the information on all services. Show each **service's total cost** (quantity × rate).

- c) Calculate the amount paid for each employee's payroll. Show the employee ID, hours worked, rate paid and amount paid (hours worked * cost per hour)

Prepared by: D. Allen Semester 1 AY 2025-26

- d) Display information from the artifacts table showing their insurance cost with tax. Compute the **insurance cost with tax** (insurance_value × 1.05).
- e) Display each player goals/match ratio. Display the player_ID, number of matches played, number of goals scored and the ratio (matches played/goals scored)
- f) Display information for each project showing the estimated cost and escalation cost. Escalation cost is the estimated cost increased by 15%.

iv) Compound Queries

- a. Write a query to extract information based on a criteria that falls within a range to be determined by you.
- b. Write a query that extracts information based on multiple criteria.
- c. Write a query that uses an IN statement.

Task 3: Date Functions (Complete the queries that are applicable to you)

i) DateDiff and Date Add (8 marks)

a. Greg's Extreme

- i. Display for each member the days remaining before their plan expires
- ii. Display for each member the type of plan, amount paid, payment date, and the expiry date for monthly plans. Expiry date is 30 days from the payment date.

b. Elegant events

- i. Display the number of days between booking and event date for all events. Assume the booking date is the current date.
- ii. Calculate the payment due date (14 days after the event date) for each client showing the client ID.

c. ZooJam

- i. For each animal placed in a cage show the duration in days for the time spent in the cage.
- ii. Display all employees who will reach retirement age in the next 8 months.

d. Collections

Prepared by: D. Allen Semester 1 AY 2025-26

- i. Display the exhibition number, theme and duration of the exhibition in days for each exhibition.
- ii. Extend exhibitions that has an a in the theme by 15 days. Display the exhibition theme and new closing date.

e. Football Federation

- i. How many days have passed since a match had a draw. Show the date of the match, the score (host, and guest) and the number of days that has passed since that match.
- ii. For each match that took place, display the return match date, the return match date is 2 weeks from the date of the current match.

f. Promaneng Company Ltd

- i. Display the number of days that each equipment is in use for each project.
 - ii. Update the project table to change the column name from estimated completion date to start date. Display for each project the project budget, ID, start date, and estimated completion date (this is 4 months after the start date)
- ii) Write a query each to demonstrate the following functions **month()**, **day()** **year()**. (6 marks)

SUBMISSION REQUIREMENTS

Your submission must include:

1. The **SQL script file (.sql)** containing all implementation steps.
2. A **screenshot or exported output** showing that all statements executed successfully.
3. A brief **reflection paragraph** describing what you learned from any two queries that

you thought were challenging.

All submissions must be uploaded by the deadline announced by your tutor/lab technologist. Late submissions will be subject to penalties as outlined in the course policy.

Prepared by: D. Allen Semester 1 AY 2025-26